

Session-10

1. Java Dequeue:

```
import java.util.*;

public class test {
    public static void main(String[] args) {
        Scanner in = new Scanner(System.in);
        Deque<Integer> deque = new ArrayDeque<>();
        HashMap<Integer, Integer> map = new HashMap<>();
        int n = in.nextInt();
        int m = in.nextInt();
        int maxUnique = 0;
        for (int i = 0; i < n; i++) {
            int num = in.nextInt();
            deque.offerLast(num);
            map.put(num, map.getOrDefault(num, 0) + 1);
            if (deque.size() > m) {
                int removed = deque.pollFirst();
                map.put(removed, map.get(removed) - 1);
                if (map.get(removed) == 0) {
                    map.remove(removed);
                }
            }
            if (deque.size() == m) {
                maxUnique = Math.max(maxUnique, map.size());
            }
        }

        System.out.println(maxUnique);
    }
}
```

Output:

The screenshot shows a challenge completion screen with a green header bar. The header contains the text "Congratulations" and "You solved this challenge. Would you like to challenge your friends?" with social media icons for Twitter and LinkedIn. A "Next Challenge" button is on the right. Below the header, there is a list of test cases on the left, each with a green checkmark and a trash icon. The test cases are labeled "Test case 0" through "Test case 6". To the right of the list, there is a "Compiler Message" section showing "Success". Below that, there is an "Input (stdin)" section showing two lines of input: "1 6 3" and "2 5 3 5 2 3 2". Below the input, there is an "Expected Output" section showing "1 3". To the right of the input and output sections, there are two "Download" buttons.

2. Java Hashset:

```
import java.io.*;
import java.util.*;
```

```

import java.text.*;
import java.math.*;
import java.util.regex.*;

public class Solution {

    public static void main(String[] args) {
        Scanner s = new Scanner(System.in);
        int t = s.nextInt();
        String [] pair_left = new String[t];
        String [] pair_right = new String[t];

        for (int i = 0; i < t; i++) {
            pair_left[i] = s.next();
            pair_right[i] = s.next();
        }

        HashSet<String> pairsSet = new HashSet<>();
        for (int i = 0; i < t; i++) {
            pairsSet.add(pair_left[i] + "," + pair_right[i]);
            System.out.println(pairsSet.size());
        }
    }
}

```

Output:

Congratulations!
 You have passed the sample test cases. Click the submit button to run your code against all the test cases.

Sample Test case 0

Input (stdin)

1 5
 2 john tom
 3 john mary
 4 john tom
 5 mary anna
 6 mary anna

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Your Output (stdout)

1 1
 2 2
 3 2